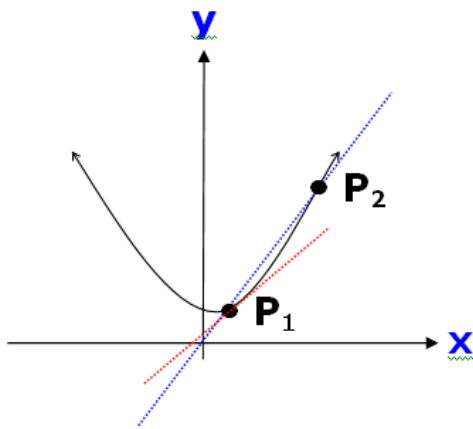


1.6: Slopes of Tangents and Instantaneous Rates of Change



What happens to the secant as P_2 gets closer to P_1 ?



Secant: a line that touches two points on a graph

Tangent: a line that touches ONE point on a graph

Compare the slope of the secant to the slope of the tangent.

➡ As P_2 gets closer to P_1 ,

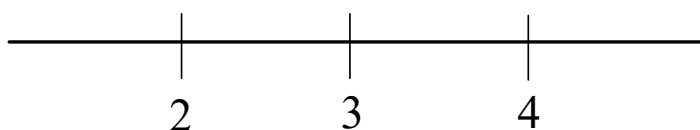
Therefore, the slope of the tangent, or, ***the Instantaneous Rate of Change***, can be approximated by:

using the equation to determine the slope of the secant using a very short interval between the tangent point and a second point on the curve (± 0.001).

Ex 1:

Given a function at the specific point $x = 3$.

As we approach this value from the left, which x -value would you use to estimate the instantaneous rate of change (the slope of the tangent)?



As we approach this value from the right, which x -value would you use to estimate the instantaneous rate of change?



Ex 2: For the f'n $f(x) = 6x^2 - 4$, estimate the Instantaneous Rate of Change when $x = -2$.

Sol'n: Use $[-2, -1.999]$
or
 $[-2.001, -2,]$

Choose a point
very close to **-2**;
to **3 dec places**

Ex 3: A diver is on a 10 m platform, preparing to dive. The diver's height above the water, in metres, at time, t , can be modeled by the equation $h(t) = 10 - 4.9t^2$.

a) Determine when the diver will **enter the water**. Round to 2 decimal places.

b) Estimate the **rate** at which the diver's height above the water is changing as the diver enters the water. Round the final answer to the nearest whole.

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REVIEW

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